

Lecture 20 — Transforming space

Fri Oct 9

Last updated: August 18, 2026.

The class outline for this day will be posted before class.

Reading: Boyd 7.1–7.2, [Lab 3](#), and Lachniet Ch. 2. Related objectives: [Lectures](#).

Learning objectives

By the end of class, students should be able to:

1. Predict the effect of scaling, reflection, shear, and rotation matrices.
2. Apply matrices to many 2D or 3D points at once.
3. Plot and compare original and transformed objects.
4. Adapt a short program that implements a transformation.
5. Interpret a matrix as a coordinate or physical-space transformation.